

Multiple Choice Questions (MCQs) on Testing in Spring

1. What is the primary focus of unit testing in Spring?

- A) Testing the entire application
 - B) Testing individual components in isolation
 - C) Testing user scenarios
 - D) Testing database interactions
- Answer:** B) Testing individual components in isolation

2. Which framework is commonly used for unit testing in Spring?

- A) JUnit
 - B) TestNG
 - C) Mockito
 - D) Both A and C
- Answer:** D) Both A and C

3. What does the @SpringBootTest annotation do?

- A) Loads only the web layer
 - B) Loads the complete application context
 - C) Creates a mock for a Spring bean
 - D) Tests JPA repositories
- Answer:** B) Loads the complete application context

4. Which annotation is used to test the web layer (controllers) in Spring?

- A) @DataJpaTest
 - B) @WebMvcTest
 - C) @SpringBootTest
 - D) @MockBean
- Answer:** B) @WebMvcTest

5. What is the purpose of the @MockBean annotation?

- A) To create a real instance of a Spring bean
 - B) To create and inject a mock for a Spring bean
 - C) To load the entire application context
 - D) To test JPA repositories
- Answer:** B) To create and inject a mock for a Spring bean

6. Which type of testing ensures that different parts of the application work together correctly?

- A) Unit Testing
 - B) Integration Testing
 - C) End-to-End Testing
 - D) Component Testing
- Answer:** B) Integration Testing

7. What is the main advantage of using Test Slices like @WebMvcTest and @DataJpaTest?

- A) They load the entire application context
- B) They speed up tests and reduce resource consumption

- C) They are used for end-to-end testing
- D) They are only for unit testing

Answer: B) They speed up tests and reduce resource consumption

8. In the context of unit testing, what does isolation mean?

- A) Tests should run in a separate environment
- B) Tests should not depend on external systems
- C) Tests should be executed in parallel
- D) Tests should cover all scenarios

Answer: B) Tests should not depend on external systems

9. Which of the following is NOT a type of testing mentioned in the document?

- A) Unit Testing
- B) Integration Testing
- C) Performance Testing
- D) End-to-End Testing

Answer: C) Performance Testing

10. What is the purpose of using MockMvc in unit testing Spring MVC controllers?

- A) To test the database interactions
- B) To test the controller in isolation
- C) To load the entire application context
- D) To perform end-to-end testing

Answer: B) To test the controller in isolation

11. Which of the following is a best practice for testing in Spring?

- A) Use real databases for unit tests
- B) Keep tests isolated
- C) Avoid using assertions
- D) Run tests manually

Answer: B) Keep tests isolated

12. What does the @DataJpaTest annotation specifically test?

- A) The web layer
- B) JPA repositories
- C) The entire application context
- D) Service classes

Answer: B) JPA repositories

13. Which of the following is a key concept for unit testing the service layer?

- A) Integration
- B) Mocking
- C) User Interface
- D) Performance

Answer: B) Mocking

14. What is the role of the TestRestTemplate in integration testing?

- A) To mock REST services
- B) To perform HTTP requests in tests
- C) To test the database

- D) To load the application context

Answer: B) To perform HTTP requests in tests

15. Which of the following is a recommended tool for asserting JSON responses in integration tests?

- A) JUnit
- B) Mockito
- C) JSONAssert
- D) AssertJ

Answer: C) JSONAssert